AQ3.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

Consider the following statements:

- Statement 1:Statement 1: $0(0,0) + 0(1,1) = (0,0)0(0,0) + 0(1,1) = (0,0)$ implies that the set $\{(0,0),(1,1)\}\{(0,0),(1,1)\}$ is a linearly independent subset of R^2 R2.
- Statement 2:Statement 2: $2(1,0) + 2(0,1) = (2,2)2(1,0) + 2(0,1) = (2,2)$ implies that the set $\{(1,0),(0,1),(2,2)\}\{(1,0),(0,1),(2,2)\}$ is a linearly dependent subset of R^2 R2.
- Statement 3:Statement 3: $2(1,0) + 2(0,1) = (2,2)2(1,0) + 2(0,1) = (2,2)$ implies that the set $\{(1,0),(2,2)\}\{(1,0),(2,2)\}$ is a linearly dependent subset of R^2 R2.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
If $\alpha(1,1,0) + \beta(0,1,1) + \gamma(1,0,1) = \upsilon\alpha(1,1,0) + \beta(0,1,1) + \gamma(1,0,1) = v$ for some $\alpha, \beta, \gamma \in R$ $\alpha, \beta, \gamma \in R$ and $\upsilon \in R^3$ $v \in R3$, then choose the set of correct options.

☐ If $\upsilon = (0,0,0)v = (0,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is $\alpha = \frac{1}{2} = \gamma$ $\alpha = 21 = \gamma$, and $\beta = -\frac{1}{2}\beta = -21$.

☐ If $\upsilon = (0,0,0)v = (0,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is $\alpha = \beta = \gamma = 0$ $\alpha = \beta = \gamma = 0$.

☐ If $\upsilon = (0,0,0)v = (0,0,0)$, then the possible solution of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is unique.

☐ If $\upsilon = (1,0,0)v = (1,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is $\alpha = \frac{1}{2} = \gamma$ $\alpha = 21 = \gamma$, and $\beta = -\frac{1}{2}\beta = -21$.

☐ The set $\{(1,1,0),(0,1,1),(1,0,1)\}\{(1,1,0),(0,1,1),(1,0,1)\}$ is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $\upsilon = (0,0,0)v = (0,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is $\alpha = \beta = \gamma = 0$ $\alpha = \beta = \gamma = 0$.
If $\upsilon = (0,0,0)v = (0,0,0)$, then the possible solution of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is unique.
If $\upsilon = (1,0,0)v = (1,0,0)$, then one of the possible solutions of $\alpha, \beta,$ and $\gamma\alpha, \beta,$ and γ is $\alpha = \frac{1}{2} = \gamma$ $\alpha = 21 = \gamma$, and $\beta = -\frac{1}{2}\beta = -21$.
The set $\{(1,1,0),(0,1,1),(1,0,1)\}\{(1,1,0),(0,1,1),(1,0,1)\}$ is linearly independent.
1 point

Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} in R^2 R2. Choose the set of correct options.

- [ ] $\{(1,2),(-1,2)\}${(1, 2), (−1, 2)} is a linearly dependent set.
- [ ] $\{(-1,2),(-3,1)\}${(−1, 2), (−3, 1)} is a linearly independent set.
- [ ] $\{(-3,1),(-1,2),(3,1)\}${(−3, 1), (−1, 2), (3, 1)} is a linearly dependent set.
- [ ] $\{(3,1),(-3,1)\}${(3, 1), (−3, 1)} is a linearly dependent set.
- [ ] $\{(3,1),(1,2),(-3,1)\}${(3, 1), (1, 2), (−3, 1)} is a linearly dependent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(-1,2),(-3,1)\}${(−1, 2), (−3, 1)} is a linearly independent set.
$\{(-3,1),(-1,2),(3,1)\}${(−3, 1), (−1, 2), (3, 1)} is a linearly dependent set.
$\{(3,1),(1,2),(-3,1)\}${(3, 1), (1, 2), (−3, 1)} is a linearly dependent set.

1 point

Let VV be a vector space and non zero vectors $v_1, v_2, v_3,$ and $v_4 \in V$v1, v2, v3, and v4 ∈ V . If v_1v1 is a linear combination of $v_i$vi, for $i = 2,3,4$i = 2, 3, 4, then which of the following is (are) correct?

- [ ] The set $\{v_1, v_2, v_3, v_4\}${v1, v2, v3, v4} cannot be linearly independent.
- [ ] The set $\{v_2, v_3, v_4\}${v2, v3, v4} must be linearly dependent.
- [ ] The set $\{v_2, v_3, v_4\}${v2, v3, v4} must be linearly independent.
- [ ] The set $\{v_1\}${v1} must be linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The set $\{v_1, v_2, v_3, v_4\}${v1, v2, v3, v4} cannot be linearly independent.
The set $\{v_1\}${v1} must be linearly independent.

1 point

Choose the correct set of options.

- [ ] Union of two distinct linearly independent sets may be linearly independent.
- [ ] Union of two distinct linearly independent sets must be linearly dependent.
- [ ] Non-empty intersection of two linearly independent sets must be linearly independent.
- [ ] Non-empty intersection of two linearly independent sets must be linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Union of two distinct linearly independent sets may be linearly independent.
Non-empty intersection of two linearly independent sets must be linearly independent.

If vectors v_1, v_2, v_3 v1, v2, v3 are linearly independent and $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$αv1 + βv2 + γv3 = 0, then find the value of $\alpha + \beta + \gamma$α + β + γ.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Consider the set of matrices $S = \{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\x\end{bmatrix}\} \subset M_{3\times 1}(R)$S = ⎩ ⎧ ⎡123⎤ , ⎡456⎤ , ⎡78x⎤ ⎭ ⎫ ⊂ M3×1(R). If $x \in \{9, 4, 2, 0\}$x ∈ {9, 4, 2, 0}, then find the maximum value of xx so that SS is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Level 2

1 point
Let SS be a linearly independent subset of R^3 R3. Choose the correct set of options.

☐ $S \cup \{(1, 0, 0)\}$S ∪ {(1, 0, 0)} must be linearly independent.

☐ $S \cup \{v\}$S ∪ {v} must be linearly independent for any $v \in R^3$ v ∈ R3.

☐ $S \setminus \{v\}$S \ {v} must be linearly independent for any $v \in R^3$ v ∈ R3.

☐ There may exist some $v \in R^3$ v ∈ R3, such that $S \cup \{v\}$S ∪ {v} is still linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$S \setminus \{v\}$S \ {v} must be linearly independent for any $v \in R^3$ v ∈ R3.
There may exist some $v \in R^3$ v ∈ R3, such that $S \cup \{v\}$S ∪ {v} is still linearly independent.
1 point
Consider a set of vectors $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$S = {(1, 1, −1), (−1, 1, 1), (0, 21, 0), (0, 1, −2), (1, 0, −2)}. Choose the set of correct options.

☐ The singleton set $\{(0, 1, -2)\}${(0, 1, −2)} is linearly dependent.

☐ If $\alpha, \beta \in S$α, β ∈ S and α, βα, β are distinct then $\{\alpha, \beta\}${α, β} is a linearly independent set of vectors.

☐ The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}${(−1, 1, 1), (0, 21, 0), (0, 1, −2)} is a linearly independent set of vectors.

☐ The set SS is a linearly independent set of vectors.

☐ The set $\{\alpha, \beta, \gamma\}${α, β, γ} is a linearly dependent set of vectors for any $\alpha, \beta, \gamma \in S$α, β, γ ∈ S, where

all the three are distinct vectors.

☐ The set $\{\alpha, \beta, \gamma, \delta\}\{\alpha, \beta, \gamma, \delta\}$ is a linearly independent set of vectors for any $\alpha, \beta, \gamma, \delta \in S$ $\alpha, \beta, \gamma, \delta \in S$, where all the four are distinct vectors.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $\alpha, \beta \in S\alpha, \beta \in S$ and $\alpha, \beta\alpha, \beta$ are distinct then $\{\alpha, \beta\}\{\alpha, \beta\}$ is a linearly independent set of vectors.
The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}\{(-1, 1, 1), (0, 21, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

1 point

Let SS be the solution set of a system of homogeneous linear equations with 3 variables and 3 equations, whose matrix representation is as follows:

$$Ax = 0Ax = 0$$

where AA is the $3 \times 33 \times 3$ coefficient matrix and xx denotes the column vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $\begin{bmatrix} \\ x1x2x3 \end{bmatrix}$.

Choose the set of correct options.

☐ If v_1 v1 and v_2 v2 are in SS, then any linear combination of v_1 v1 and v_2 v2 will also be in SS.

☐ The set SS will be a subspace of R^3 R3, with respect to usual addition and scalar multiplication as in R^3 R3.

☐ The set $\{v_1, v_2, v_1 - v_2\}\{v1, v2, v1 - v2\}$ is a linearly dependent subset in SS.

☐ The set $\{v_1, v_2\}\{v1, v2\}$ is a linearly independent subset in SS if v_1 v1 is not a scalar multiple of v_2 v2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If v_1 v1 and v_2 v2 are in SS, then any linear combination of v_1 v1 and v_2 v2 will also be in SS.
The set SS will be a subspace of R^3 R3, with respect to usual addition and scalar multiplication as in R^3 R3.
The set $\{v_1, v_2, v_1 - v_2\}\{v1, v2, v1 - v2\}$ is a linearly dependent subset in SS.
The set $\{v_1, v_2\}\{v1, v2\}$ is a linearly independent subset in SS if v_1 v1 is not a scalar multiple of v_2 v2.

Check Answers
Your score is: 0/10